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3. Water Jug Problem

AIM

To write a Python program to solve the Water Jug Problem using Breadth First Search (BFS).

PSEUDO CODE

1. Start
2. Define jug capacities and target
3. Initialize queue with empty jugs
4. Generate all possible valid states
5. Check whether target is reached
6. Display solution path
7. Stop

PROGRAM

```
from collections import deque

cap1, cap2 = 4, 3
target = 2
start = (0,0)

def next_states(a,b):
    return {
        (cap1,b), (a,cap2), (0,b), (a,0),
        (max(0,a-(cap2-b)), min(cap2,a+b)),
        (min(cap1,a+b), max(0,b-(cap1-a)))
    }

q=deque([(start,[])])
visited={start}

while q:
    (a,b),path=q.popleft()
    if a==target or b==target:
        print(path+[(a,b)])
        break
    for s in next_states(a,b):
        if s not in visited:
```

```
visited.add(s)
q.append((s,path+[(a,b)]))
```

INPUT

Jug 1 Capacity = 4

Jug 2 Capacity = 3

Target = 2 litres

OUTPUT

[(0,0), (4,0), (1,3), (1,0), (0,1), (4,1), (2,3)]

RESULT

The Python program for the Water Jug Problem was executed successfully and the target quantity was obtained.

4. Crypt-Arithmetic Problem

AIM

To write a Python program to solve the Crypt-Arithmetic problem (SEND + MORE = MONEY).

PSEUDO CODE

8. Start
9. Assign unique digits to letters
10. Ensure leading letters are not zero
11. Form numbers SEND, MORE and MONEY
12. Check SEND + MORE = MONEY
13. Display solution
14. Stop

PROGRAM

```
from itertools import permutations

letters='SENDMORY'
digits='0123456789'

for p in permutations(digits,8):
    m=dict(zip(letters,p))
    if m['S']=='0' or m['M']=='0':
        continue
    send=int(m['S']+m['E']+m['N']+m['D'])
    more=int(m['M']+m['O']+m['R']+m['E'])
    money=int(m['M']+m['O']+m['N']+m['E']+m['Y'])
    if send+more==money:
        print(send, more, money)
        break
```

INPUT

No input required

OUTPUT

$9567 + 1085 = 10652$

RESULT

The Python program for the Crypt-Arithmetic problem was executed successfully and the correct solution was obtained.